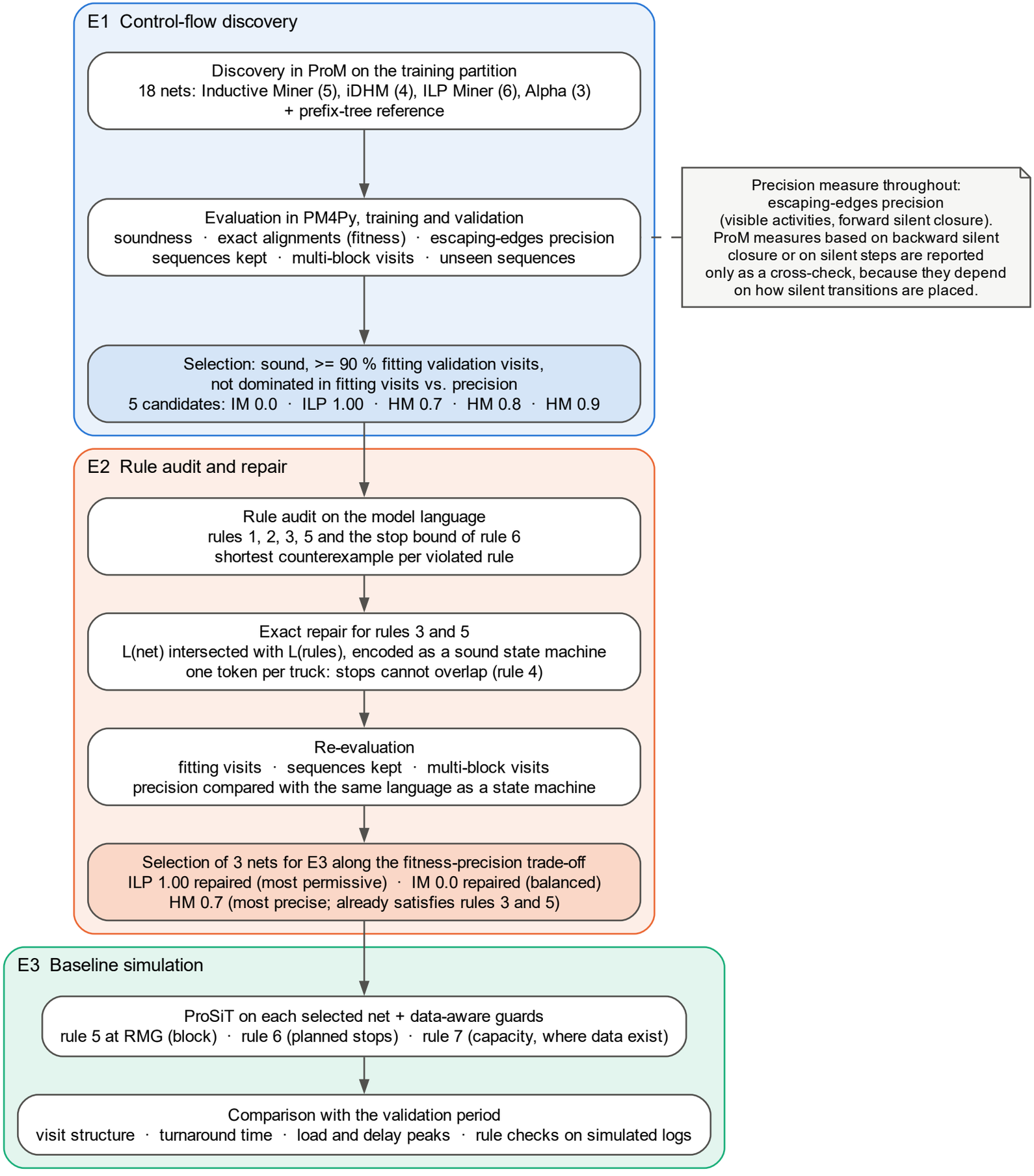
E1 Control-flow discovery
Discovery in ProM on the training partition
18 nets: Inductive Miner (5), iDHM (4), ILP Miner (6), Alpha (3)
+ prefix-tree reference
Evaluation in PM4Py, training and validation
soundness · exact alignments (fitness) · escaping-edges precision
sequences kept · multi-block visits · unseen sequences
Selection: sound, >= 90 % fitting validation visits,
not dominated in fitting visits vs. precision
5 candidates: IM 0.0 · ILP 1.00 · HM 0.7 · HM 0.8 · HM 0.9
Precision measure throughout:
escaping-edges precision
(visible activities, forward silent closure).
ProM measures based on backward silent
closure or on silent steps are reported
only as a cross-check, because they depend
on how silent transitions are placed.
E2 Rule audit and repair
Rule audit on the model language
rules 1, 2, 3, 5 and the stop bound of rule 6
shortest counterexample per violated rule
Exact repair for rules 3 and 5
L(net) intersected with L(rules), encoded as a sound state machine
one token per truck: stops cannot overlap (rule 4)
Re-evaluation
fitting visits · sequences kept · multi-block visits
precision compared with the same language as a state machine
Selection of 3 nets for E3 along the fitness-precision trade-off
ILP 1.00 repaired (most permissive) · IM 0.0 repaired (balanced)
HM 0.7 (most precise; already satisfies rules 3 and 5)
E3 Baseline simulation
ProSiT on each selected net + data-aware guards
rule 5 at RMG (block) · rule 6 (planned stops) · rule 7 (capacity, where data exist)
Comparison with the validation period
visit structure · turnaround time · load and delay peaks · rule checks on simulated logs